



PROJECT TEAM

P. Sai Medhas (VTU24914) · J. Rithik Reddy (VTU25092) · A. Kaif (VTU25108)

ACADEMIC SUPERVISOR

Dr. S. Gokul Pran — Asst. Professor (Sr. Grade)

INSTITUTION

Vel Tech Ranganathan Dr. Sagunthala R&D Institute of Science and Technology, Chennai — Dept. of CSE

1 EXECUTIVE SUMMARY

⚠️ BOTTLENECK

Manual auditing in emergency healthcare causes unexpected stockouts mid-trauma response, unnoticed expiry dates, high medical waste, and zero real-time visibility into kit contents.

✓ SOLUTION

A battery-powered **smart medical container** fusing multi-sensor edge telemetry (ESP32), MQTT/HTTP cloud sync, and predictive AI to automate stock monitoring, expiry alerts & reordering.

2 5-LAYER SYSTEM ARCHITECTURE

APPLICATION LAYER

Web Dashboard

Mobile App

Low-Stock Alerts

Supplier Notification

AI & PREDICTIVE ANALYTICS LAYER

Usage Pattern Analysis

Demand Forecasting

Shelf-Life Decay

Auto-Restock Suggestion

CLOUD & STORAGE LAYER

Firebase / AWS IoT Core

Time-Stamped Telemetry

Batch Metadata & Unit IDs

IOT COMMUNICATION LAYER

ESP32 Wi-Fi / GSM

MQTT

HTTP REST

Encrypted JSON Payloads

HARDWARE / EDGE LAYER

ESP32 MCU

RC522 RFID + Tags

Load Cell + HX711 ADC

0.96" OLED I2C

Piezo Buzzer

Figure 1 — 5-Layer IoT & AI System Architecture

3 EDGE HARDWARE COMPONENT ROLES

COMPONENT

ROLE IN SYSTEM

ESP32 MCU

Central edge controller — event processing, sensor polling & cloud sync.

RC522 RFID

Reads unique tag IDs on each medical item for identification on withdrawal.

Load Cell + HX711

24-bit ADC amplifier for continuous weight/quantity verification.

OLED + Buzzer

0.96" I2C display + active piezo buzzer for on-kit visual/audio alerts.

4 AUTONOMOUS LOGIC PIPELINE

A Item Withdrawal

RFID tag identification fires alongside a load-cell weight-delta detection for dual-mode confirmation.

B ESP32 Event Processing

Edge controller validates the event and packages an encrypted JSON telemetry payload.

C Real-Time Cloud Sync

Payload streamed via MQTT/HTTP to Firebase / AWS IoT Core, updating stamped inventory logs.

D AI Consumption Analysis

ML pipeline tracks consumption velocity and shelf-life decay to predict restocking thresholds.

E Automated Alerts & Restock Trigger

On-kit visual/audio alert fires and a supplier dispatch hook is triggered for low stock or expiry.

5 ML EVALUATION & LATENCY BENCHMARK

92%

ACCURACY

90%

PRECISION

88%

RECALL

89%

F1-SCORE

0.91

ROC-AUC

METRIC

VALUE

Time-Series Demand Error (RMSE / MAE)

2.5 / 1.8

Telemetry Update Latency

< 2 seconds

Alert Response Latency

~ 1 second

System Uptime & Reliability

95%

6 PHYSICAL PROTOTYPE — LABELED COMPONENTS

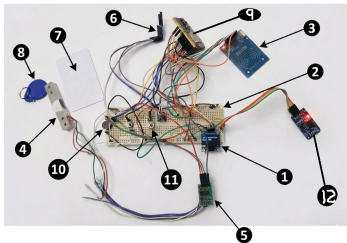


Figure 2 — Edge prototype with numbered components on breadboard

9 ESP32 Microcontroller (MCU)

4 Load Cell (weight sensor)

1 0.96" OLED I2C Display

6 Wi-Fi/GSM Communication Module

7/8 RFID Tag Card & Keyfob

3 RC522 RFID Reader

5 HX711 Load Cell Amplifier

12 Active Piezo Buzzer Module

2 Breadboard

11 Push Button (manual trigger)

7 ENGINEERING HIGHLIGHTS

01

Dual-mode verification — RFID + load-cell weight sensing together prevent unrecorded item removals.

02

Low-power, battery-efficient design built specifically for mobile ambulance and field environments.

03

Standalone resilience with edge failover and seamless cloud re-synchronization on reconnect.